

Input Requirements and Processing Logic

This document describes the mandatory input columns, identification (ID) hierarchy, and comparison handling for the Shiny app. It is intended to clarify what is required for each supported tool and how the app processes the uploaded data.

1. General Input File Structure

The app expects a **xlsx file** exported from Compound Discoverer 3.5 (CD) with original column names (e.g. Log2 Fold Change, P-value, etc.). Each tool requires different input columns stated in 2. Decide, how strict you want to filter your results beforehand based on different annotation levels. (mzVault/in-house database, mzCloud/spectral library, MassList, ChemSpider, LIPIDMaps, etc.)

Note: Each comparison is treated by the app.

The file for the functional analysis in MetaboAnalyst can be exported directly from CD as it requires all compounds. Depending on the input option, different columns are required (see UserGuide MetaboAnalyst). The column names need to be adjusted. In addition, the mode column needs to be translated into positive or negative when selecting the “Reference Ion” column from CD.

2. Mandatory Input Columns per Tool

The following table summarizes which identifier columns are **obligatory** for each tool to run.

Note:

- All tools require at least one **P-value or Adj. P-value** column.
- KEGG and MetScape require all metabolites despite if they are significant or not
- The other tools filter significant features before exporting them but work on prefiltered data if all the required columns are present.
- MetaboAnalyst and RaMP-DB need **at least one** of the **ID** columns listed
- A notification pops up if a column for a specific tool is missing.

Table 1: List of required columns for each tool that can be selected:

Tool	ID column	Statistics	Optional
MetaboAnalyst	HMDB ID KEGG ID	Log2 FC (Adj.) P-value	Name, m/z, RT
RaMP-DB	HMDB ID KEGG ID CSID	Log2 FC (Adj.) P-value	Name, m/z, RT
GOA	InChIKey	Log2 FC (Adj.) P-value	Name, m/z, RT
KEGG	KEGG ID	Log2 FC (Adj.) P-value	Name, m/z, RT, (norm.) Area
FELLA	KEGG ID	Log2 FC (Adj.) P-value	Name, m/z, RT, (norm.) Area
MetScape	KEGG ID	Ratio (Adj.) P-value	Name, m/z, RT

3. ID Selection Hierarchy

When multiple identifier columns are present in the input file, the app applies a **fixed priority order** to determine which ID is used for downstream analysis. If no valid identifier can be resolved for a tool, that compound is excluded from the respective tool run.

Files from VBCF GmbH: compounds from our in-house database (ML) are always treated as a priority as these are level 1 identified (m/z, RT, and MS/MS).

Duplicate IDs are avoided; if an ID has already been used, it is skipped. This is especially important when ML and mzCloud hits are listed in one file (VBCF files only).

When MetaboAnalyst or RaMPDB tools are selected the hierarchy is:

1. Internal database hits (ML)

1. HMDB ML
2. KEGG ML
3. (ChemSpider ML RaMPDB only)

2. mzClouds hits

1. HMDB ID
2. KEGG ID

4. Handling of Multiple Comparisons

If the uploaded file contains **comparisons between multiple groups or conditions**, the app will:

- Detect all valid log2FC/p-value column pairs
- Run each selected tool independently per comparison
- Generate a separate output file for each comparison with the specific name

This means:

- Each tool produces its own result folder
- Each comparison produces its own file within the result folder
- Downstream tools never mix results from different comparisons

5. Notes on MetaboAnalyst and RaMPDB ID Selection

For MetaboAnalyst and RaMP-DB, the current implementation uses the fixed ID hierarchy described above.

Future development:

- Selection of additional or alternative ID types for these tools
- User-defined control over which identifier column is used

6. Quick Checklist

Before running the Shiny app, make sure the following requirements are met:

Input File

- ☐ File is directly exported from CD 3.5 as **xslx**
- ☐ Contains at least one valid **compound ID column**
- ☐ Contains **log2FC and (adj.) p-value columns** for at least one comparison

Parameters

- ☐ Thresholds (log2FCs, (adj.) p-value) are chosen according to your biological question

7. Summary

- Each tool has defined mandatory identifier requirements
- A strict ID hierarchy ensures consistent and reproducible mapping
- Each comparison is processed independently
- log2FC and p-values control filtering and interpretation
- Extended ID flexibility for MetaboAnalyst and RaMP-DB is planned